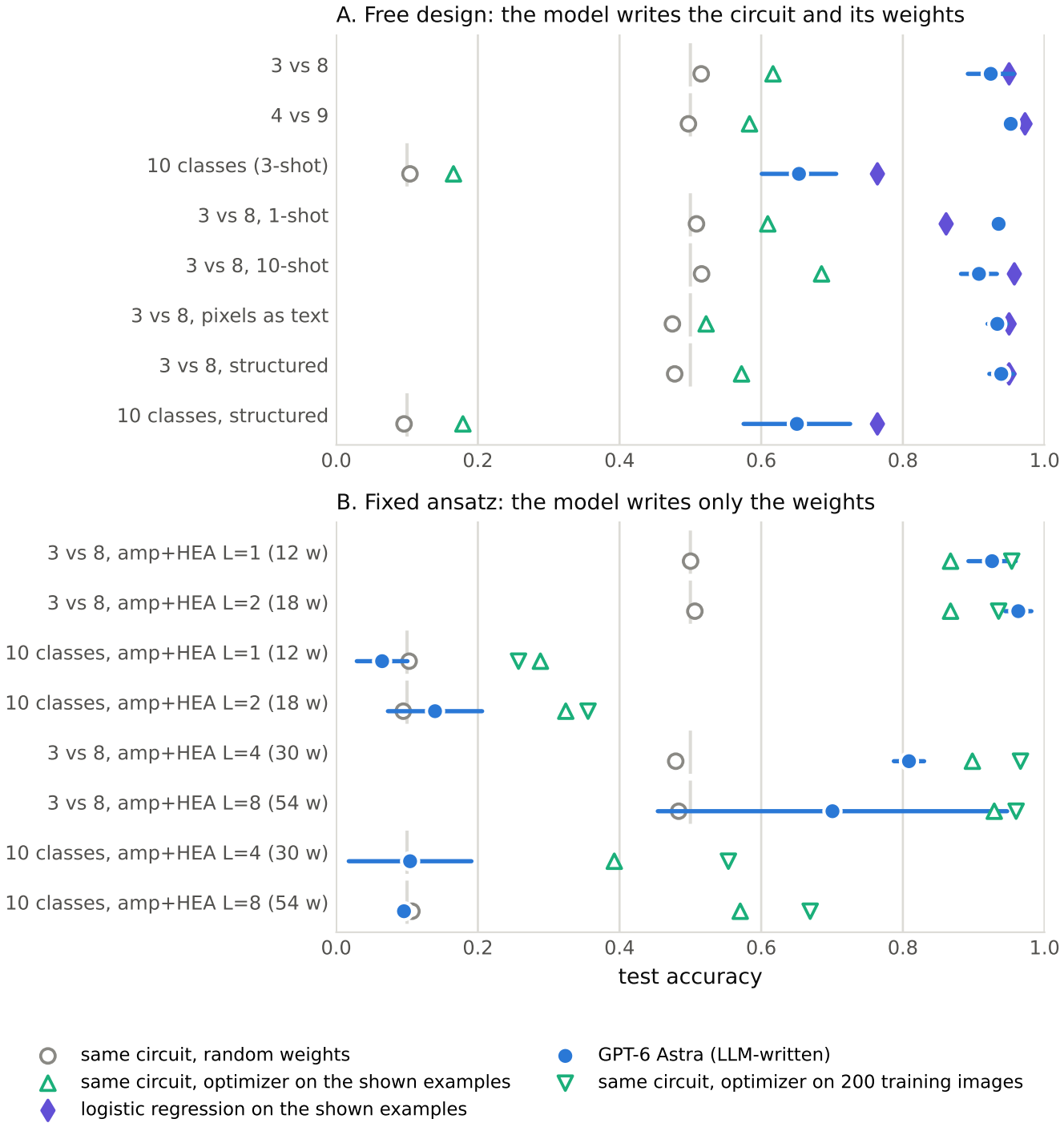
A. Free design: the model writes the circuit and its weights
3 vs 8
4 vs 9
10 classes (3-shot)
3 vs 8, 1-shot
3 vs 8, 10-shot
3 vs 8, pixels as text
3 vs 8, structured
10 classes, structured
0.0
0.2
0.4
0.6
0.8
1.0
B. Fixed ansatz: the model writes only the weights
3 vs 8, amp+HEA L=1 (12 w)
3 vs 8, amp+HEA L=2 (18 w)
10 classes, amp+HEA L=1 (12 w)
10 classes, amp+HEA L=2 (18 w)
3 vs 8, amp+HEA L=4 (30 w)
3 vs 8, amp+HEA L=8 (54 w)
10 classes, amp+HEA L=4 (30 w)
10 classes, amp+HEA L=8 (54 w)
0.0
0.2
0.4
0.6
0.8
1.0
test accuracy
same circuit, random weights
same circuit, optimizer on the shown examples
logistic regression on the shown examples
GPT-6 Astra (LLM-written)
same circuit, optimizer on 200 training images